

CAEN CARPIQUET, France

WMO: 070270

Lat: 49.1800N

Lon: 0.4561W

Elev: 78

StdP: 100.39

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.6	-2.0	-6.7	2.2	-2.0	-4.9	2.5	-1.0	15.0	10.1	13.7	10.3	3.0	130	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.7	28.0	19.8	25.7	18.9	23.8	18.1	20.7	26.5	19.6	24.4	18.6	22.7	4.2	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.5	23.0	17.7	12.8	21.9	16.9	12.2	20.8	59.7	26.8	56.1	24.4	53.0	22.7	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
11.9	10.4	9.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-6.2	32.9	2.5	2.6	-8.0	34.7	-9.5	36.3	-10.9	37.7	-12.8	39.6		
			-6.8	22.8	2.5	1.1	-8.6	23.7	-10.0	24.3	-11.4	25.0	-13.2	25.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	5.5	5.9	7.7	9.7	12.7	15.6	17.8	17.9	15.5	12.6	8.6	6.0
	DBStd	5.39	3.73	3.32	2.97	2.85	2.79	2.74	2.45	2.60	2.54	3.01	3.04	3.73
	HDD10.0	593	144	118	82	39	7	0	0	0	1	12	59	131
	HDD18.3	2626	399	348	329	259	175	89	39	39	92	179	292	383
	CDD10.0	1074	3	3	12	29	91	169	242	245	164	91	18	6
	CDD18.3	66	0	0	0	0	1	9	23	26	6	1	0	0
	CDH23.3	546	0	0	0	2	17	101	177	193	50	5	0	0
	CDH26.7	142	0	0	0	0	2	31	42	58	9	1	0	0
Wind	WSAvg	4.4	5.3	5.1	4.7	4.3	4.0	3.8	3.9	3.9	3.9	4.3	4.7	5.0
Precipitation	PrecAvg	761	69	58	52	55	60	57	57	58	56	77	82	79
	PrecMax	1003	167	135	167	117	116	159	135	131	126	162	144	199
	PrecMin	512	10	10	7	6	13	10	11	9	7	5	21	9
	PrecStd	110	35	32	34	31	28	34	31	37	29	36	34	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	15.3	18.7	22.9	25.6	30.6	30.9	31.8	27.9	23.4	17.2	14.4
		MCWB	12.0	11.0	12.6	15.8	18.2	20.9	20.6	20.7	18.8	17.7	15.0	12.7
	2%	DB	12.8	13.0	15.9	19.2	22.3	26.1	27.2	27.9	24.4	20.1	15.3	13.4
		MCWB	11.4	10.6	11.9	13.7	16.6	19.1	19.6	19.5	17.9	16.4	13.4	12.0
	5%	DB	11.7	11.8	14.1	16.8	20.2	23.1	25.1	25.1	22.0	18.3	14.2	12.4
		MCWB	10.3	9.9	11.2	12.4	15.6	18.0	18.7	18.8	17.0	15.4	12.5	11.1
	10%	DB	10.6	10.8	12.6	14.8	18.2	21.1	23.2	23.0	20.2	17.0	13.1	11.4
		MCWB	9.2	9.1	10.2	11.2	14.3	16.7	17.8	17.7	16.0	14.7	11.5	10.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	12.0	13.8	16.2	19.2	22.1	21.8	21.9	20.2	18.2	15.5	13.2
		MCDB	13.5	14.0	17.2	21.2	24.1	28.4	28.9	29.0	25.2	21.7	16.8	14.1
	2%	WB	11.5	11.0	12.5	14.2	17.4	19.9	20.3	20.5	18.6	17.0	13.8	12.2
		MCDB	12.6	12.6	15.0	18.5	21.4	25.2	25.9	26.4	22.9	19.5	15.1	13.2
	5%	WB	10.4	10.2	11.5	12.9	15.9	18.3	19.2	19.4	17.6	15.9	12.7	11.3
		MCDB	11.5	11.6	13.7	16.0	19.5	22.4	23.8	23.9	21.0	17.8	14.0	12.4
	10%	WB	9.3	9.2	10.5	11.7	14.7	17.1	18.3	18.4	16.7	15.0	11.8	10.2
		MCDB	10.4	10.7	12.4	14.2	17.7	20.5	22.3	22.0	19.6	16.7	13.0	11.3
Mean Daily Temperature Range	5% DB	MDBR	5.3	6.1	7.4	8.6	8.8	9.4	9.7	9.7	9.1	7.5	6.0	5.3
		MCDBR	5.6	7.5	10.0	12.3	12.5	13.4	13.4	13.2	12.1	9.1	6.6	5.5
		MCWBR	4.9	5.6	6.7	7.6	7.5	7.6	6.6	6.2	6.5	5.8	5.2	4.8
	5% WB	MCDBR	5.5	6.6	8.9	10.8	11.3	12.3	11.9	11.6	10.5	8.2	6.0	5.4
		MCWBR	5.1	5.4	6.2	7.0	7.3	7.4	6.4	6.0	6.1	5.7	5.1	5.0
Clear-Sky Solar Irradiance	taub	0.326	0.343	0.376	0.394	0.400	0.399	0.393	0.387	0.371	0.357	0.333	0.320	
	taud	2.425	2.392	2.295	2.260	2.284	2.322	2.341	2.379	2.407	2.441	2.443	2.394	
	Ebn at Noon	725	800	827	851	860	862	861	850	827	773	710	674	
	Edh at Noon	65	84	108	124	126	123	119	109	96	79	63	59	
All-Sky Solar Radiation	RadAvg	0.92	1.66	2.83	4.40	5.19	5.78	5.56	4.66	3.61	2.04	1.11	0.76	
	RadStd	0.08	0.20	0.31	0.42	0.42	0.58	0.37	0.34	0.24	0.17	0.13	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page